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Process unit including first, second and third conveyance members for conveying developer

Abstract

A cartridge including a rotatable image bearing member which bears a developer, a cleaning member for removing a developer remaining on a surface of the image bearing member, the cleaning member having a contact portion in contact with the surface of the image bearing member and a support portion which supports the contact portion, a developer recovering portion which recovers the developer removed by the cleaning member, a first conveyance member which is disposed inside the developer recovering portion and conveys the recovered developer in a rotation axis direction of the image bearing member, a second conveyance member which conveys the developer conveyed by the first conveyance member in a direction crossing the rotation axis direction toward a discharge port provided in the developer recovering portion, and a third conveyance member to convey the developer toward the center of the photosensitive drum in the rotational axis direction. The second conveyance member is disposed at an inner side of the contact portion of the cleaning member with respect to an end part of the contact portion of the cleaning member.

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention

(1) The present invention relates to a cartridge.
Description of the Related Art
(2) An image forming apparatus which forms an image on a recording medium by using electrophotography is known. Examples of the electrophotographic image forming apparatus include an electrophotographic copying machine, an electrophotographic printer (an LED printer, a laser-beam printer, and the like), a facsimile device, a word processor and the like.
(3) Various cartridges are configured to be detachably attached to an apparatus main body of such the image forming apparatus as above. Examples of the cartridges include a development cartridge including development means and a process cartridge including the development means and a photosensitive drum. The cartridges refer to those typically having at least one of a developer, a photosensitive drum, and process means acting on the photosensitive drum and capable of being detachably attached to an image forming apparatus main body (printer main body). As a typical example of the cartridge, the process cartridge can be cited. This process cartridge is configured such that an image bearing member and process means acting on the image bearing member are made into a cartridge and is detachably attached to the apparatus main body. Moreover, a toner cartridge which supplies a toner to the process cartridge or the development cartridge is known.
(4) Japanese Patent Application Publication No. 2012-208167 proposes such a configuration having a first screw conveyance portion which conveys a waste toner on a photosensitive drum which has been scraped by a cleaning blade in an axial direction of the photosensitive drum and a second screw conveyance portion which conveys the waste toner received from the first screw conveyance portion in a direction different from the axial direction of the photosensitive drum in a region not crossing the photosensitive drum in the axial direction of the photosensitive drum and supplies it to a waste toner box.
SUMMARY OF THE INVENTION
(5) The present invention was made in view of the prior art described above and an object thereof is to develop an art in an image forming apparatus having such a configuration that the toner is conveyed by rotation of a conveyance member.
(6) The present invention provides a cartridge comprising: a rotatable image bearing member configured to bear a developer; a cleaning member configured to remove the developer remaining on a surface of the image bearing member, the cleaning member including a contact portion in contact with a surface of the image bearing member and a support portion configured to support the contact portion; a developer recovering portion configured to recover the developer removed by the cleaning member; a first conveyance member disposed inside the developer recovering portion and configured to convey the recovered developer in a rotation axis direction of the image bearing member; and a second conveyance member configured to convey the developer conveyed by the first conveyance member in a direction crossing the rotation axis direction toward a discharge port provided in the developer recovering portion, wherein in the rotation axis direction, the second conveyance member is disposed in an inner side of the contact portion of the cleaning member with respect to an end part of the contact portion of the cleaning member.
(7) According to the present invention, the art in the image forming apparatus having the configuration that the toner is conveyed by rotation of the conveyance member can be developed.
(8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a sectional view illustrating a schematic configuration of a laser printer;
(2) FIG. **2** is a front view illustrating an outline of a process cartridge;

(3) FIG. **3** is a sectional view illustrating a schematic configuration of the process cartridge;
(4) FIG. **4** is a sectional view illustrating a second toner conveyance path of the process cartridge;
(5) FIG. **5** is a section illustrating a replenishing port of the process cartridge;
(6) FIG. **6** is a front view illustrating an outline of a toner cartridge;
(7) FIG. **7** is a sectional view illustrating a toner supply portion of the toner cartridge;
(8) FIG. **8** is a sectional view illustrating a waste-toner recovering portion of the toner cartridge:
(9) FIG. **9**A is an exploded perspective view of the process cartridge;
(10) FIG. **9**B is an exploded perspective view in a different direction of the process cartridge;
(11) FIG. **10**A is a schematic side view illustrating contact of a development unit with a photosensitive drum;
(12) FIG. **10**B is a schematic side view illustrating separation of the development unit from the photosensitive drum;
(13) FIGS. **11**A and **11**B are schematic perspective views illustrating attachment of the process cartridge and the toner cartridge:
(14) FIGS. **12**A to **12**C are schematic side views illustrating attachment of the process cartridge and the toner cartridge;
(15) FIG. **13**A is an exploded perspective view of the toner cartridge:
(16) FIG. **13**B is an exploded perspective view in a different direction of the toner cartridge;
(17) FIG. **14** is a sectional perspective view illustrating a replenishing port of the process cartridge:
(18) FIGS. **15**A and **15**B are schematic sectional views illustrating a relationship between a toner accepting chamber and a stay at contact and separation positions;
(19) FIG. **16** is a schematic perspective view of the toner accepting chamber of the process cartridge;
(20) FIG. **17** is a schematic sectional view when a cleaning unit is seen from an upper surface;
(21) FIG. **18** is a schematic perspective view of a drive row of the process cartridge;
(22) FIG. **19** is a sectional view illustrating an outline of a process cartridge in Embodiment 2;
(23) FIG. **20** is a sectional view illustrating a second toner conveyance path of the process cartridge in Embodiment 2;
(24) FIG. **21** is a schematic sectional view when the cleaning unit is seen from an upper surface;
(25) FIG. **22** is a schematic perspective view of the drive row of the process cartridge;
(26) FIG. **23** is a schematic sectional view for explaining a positional relationship between the cleaning blade and the second conveyance path:
(27) FIG. **24** is a schematic sectional view of a positional relationship between a first conveyance member and a second conveyance member;
(28) FIG. **25** is a schematic sectional view passing through axes of the first conveyance member and the second conveyance member:
(29) FIG. **26** is a schematic sectional view when the cleaning unit is seen from the upper surface;
(30) FIG. **27** is a schematic perspective view of the drive row of the process cartridge;
(31) FIG. **28** is a schematic sectional view illustrating another configuration for conveyance;
(32) FIG. **29** is a front view of the conveyance member with another configuration for the conveyance; and
(33) FIGS. **30**A and **30**B are sectional views of a third conveyance member with another configuration for the conveyance.

DESCRIPTION OF THE EMBODIMENTS

Embodiment 1

(34) Hereinafter, Embodiments for working the present invention will be explained in exemplification in detail. However, dimensions, materials, shapes, relative dispositions thereof and the like of components described in the following Embodiments should be changed as appropriate in accordance with configurations and various conditions of apparatuses to which the present invention is applied. Therefore, the scope of the present invention is not limited to that unless

otherwise specifically described.

(35) Overall Outline of Laser Printer

(36) FIG. **1** is a sectional view illustrating a schematic configuration of a laser printer, which is an example of an image forming apparatus. As shown in FIG. **1**, the laser printer **1** is configured by including a printer main body P, a process cartridge B, and a toner cartridge C. In the printer main body P, a sheet feeding portion **103**, a transfer roller **104**, a fixing portion **105**, and a laser scanner **101** are installed. Moreover, in the printer main body P, the process cartridge B and the toner cartridge C are detachably disposed.

(37) The process cartridge B will be explained by using FIGS. **2**, **3**, **4**, and **5**. FIG. **2** is a front view illustrating an outline of the process cartridge B. FIG. **3** is a sectional view illustrating a schematic configuration of the process cartridge B (a-a section in FIG. **2**). FIG. **4** is a sectional view (b-b section in FIG. **2**) illustrating a second toner conveyance path **10***c* of the process cartridge B. FIG. **5** is a sectional view (c-c section in FIG. **2**) illustrating a replenishing port of the process cartridge B.

(38) As shown in FIGS. **2**, **3**, **4**, and **5**, the process cartridge B is constituted by a cleaning unit **10** (first unit) including a photosensitive drum **11** (image bearing member) and a development unit **15** (second unit) including a development roller **16** as development means for bearing a developer (toner).

(39) The cleaning unit **10** has the photosensitive drum **11** described above, a cleaning blade **17** as a cleaning member of the photosensitive drum **11**, a charging roller **12** as a charging member, a charging roller cleaner **14** as a cleaning member of the charging roller **12**, a waste-toner accommodating portion **10***a*, and the second toner conveyance path **10***c*.

(40) The charging roller **12** is disposed so as to be in contact with an outer peripheral surface of the photosensitive drum **11** and charges the photosensitive drum **11** by voltage application from the printer main body P. Moreover, the charging roller **12** is driven and rotated by the photosensitive drum **11**.

(41) The cleaning blade **17** is a member having elasticity disposed so as to be in contact with the outer peripheral surface of the photosensitive drum **11**. The cleaning blade **17** is constituted by a blade portion **17***a* (contact portion) constituted by an elastic member and a support member **17***b* (support portion). The cleaning blade **17** removes the toner remaining from the photosensitive drum **11** by a distal end thereof elastically in contact with the photosensitive drum **11**, after a sheet S, which will be described later, passes between the photosensitive drum **11** and the transfer roller **104**. The removed tonner (waste toner) is conveyed from the waste-toner accommodating portion **10***a*, which will be described later, to the toner cartridge C through a first toner conveyance path **10***b* and the second toner conveyance path **10***c*.

(42) As shown in FIG. **5**, the development unit **15** has a development chamber **151** in which the development roller **16** is disposed, a developer accommodating chamber **152** which supplies a toner to the development chamber **151**, and a toner accepting chamber **153** which accepts the toner supplied from the toner cartridge C.

(43) The development roller **16** supplies a toner to a development region of the photosensitive drum **11**. Then, the development roller **16** develops an electrostatic latent image formed on the photosensitive drum **11** by using the toner. The development blade **18** is brought into contact with a peripheral surface of the development roller **16** and regulates an amount of the toner adhering to the peripheral surface of the development roller **16**. Moreover, it applies a triboelectric charge to the toner.

(44) The toner accommodated in the developer accommodating chamber **152** is sent out to the development chamber **151** by rotation of a stirring member **154** and is supplied to the development roller **16**. A remaining amount of the toner in the developer accommodating chamber **152** is detected by remaining-amount detecting means, not shown, and when the toner amount in the developer accommodating chamber **152** becomes certain or less, the toner is supplied from the

toner cartridge C to the process cartridge B. The toner having been supplied to the development unit **15** through a replenishing port **21***c* and a delivery port **21***d* of the stay **21** is supplied to the developer accommodating chamber **152** through the accepting chamber **153**.
(45) Though details will be described later, the process cartridge B and the toner cartridge C can be detachably attached to the printer main body P.
(46) Subsequently, an operation of the laser printer **1** will be explained by using FIG. **1**. The photosensitive drum **11** rotated and driven by a drive source, not shown, is uniformly charged by the charging roller **12** to a predetermined potential. A surface of the photosensitive drum **11** after the charging is exposed by the laser scanner **101** on the basis of image information, charge on an exposed part is removed, and the electrostatic latent image is formed. The toner is supplied from the development roller **16** to the electrostatic latent image on the photosensitive drum **11** and is visualized as a toner image.
(47) On the other hand, in parallel with a forming operation of the toner image as above, the sheet S is conveyed along the sheet feeding portion **103**. Specifically, a feeding roller **103***b* is rotated and feeds the sheet S. After that, the sheet S is conveyed to a nip portion between the photosensitive drum **11** and the transfer roller **104** by matching timing with image forming of the toner image on the photosensitive drum **11**. When passing through the nip portion, the toner image is transferred as an unfixed image on the sheet S by bias application to the transfer roller **104**. After that, the sheet S on which the toner image has been transferred is conveyed to the fixing portion **105**. When the sheet S having been conveyed to the fixing portion **105** passes through the fixing portion **105**, the unfixed image is heated and pressurized and fixed on the surface of the sheet S. Moreover, it is conveyed by the sheet feeding portion **103**, ejected to an ejection tray **106** and stacked.
(48) Outline of Process Cartridge B
(49) By using FIGS. **3**, **9**A, **9**B, **10**A, and **10**B, the configuration of the process cartridge B in this Embodiment will be explained in detail. FIG. **9**A is an exploded perspective view of the process cartridge B seen from a side (one end side) where drive from the printer main body P is transmitted to the process cartridge B. FIG. **9**B is an exploded perspective view of the process cartridge B seen from a side opposite to FIG. **9**A (the other end side) with respect to the axial direction of the photosensitive drum **11**. FIG. **10**A is a schematic side view illustrating contact of the development unit **15** with respect to the photosensitive drum **11** in the process cartridge. FIG. **10**B is a schematic side view illustrating separation of the development unit **15** with respect to the photosensitive drum **11** in the process cartridge.
(50) As described above, the cleaning unit **10** has the photosensitive drum **11**, the charging roller **12**, and the cleaning blade **17**. Similarly, the development unit **15** has the development roller **16**, the development blade **18**, the development chamber **151**, the developer accommodating chamber **152**, and the toner accepting chamber **153**.
(51) As shown in FIGS. **9**A and **9**B, bearing members **4**, **5** are disposed on an end part in the axial direction of the development roller **16**, and the development unit **15** is connected to the cleaning unit **10** rotatably around a swing axis **8** defined by a straight line including a spindle **8***a* and a spindle **8***b*. The swing axis **8** is disposed substantially in parallel with a rotation axis **11***b* of the photosensitive drum **11**.
(52) A configuration in which the development unit **15** is supported by the cleaning unit **10** will be explained in detail.
(53) As shown in FIG. **9**A, a gear member **5***a* provided on the bearing member **5** is supported by a cylindrical hole shape **7***a* provided in a side cover **7** of the cleaning unit **10**. The spindle **8***a* is defined by a common axis of the cylindrical hole shape **7***a* of the side cover **7** and the gear member **5***a*. Moreover, as shown in FIG. **9**B, a pin **6** is inserted so as to go across a cylindrical hole shape **20***a* of a cleaning frame body **20** and the cylindrical hole shape **4***a* of the bearing member **4**. The spindle **8***b* is defined by a common axis of the pin **6** and the cylindrical hole shape **4***a* of the bearing member **4**. The spindle **8***a* and the spindle **8***b* are disposed substantially coaxially, and the

swing axis **8** is defined by a straight line including the spindle **8***a* and the spindle **8***b* as described above.

(54) As described above, the development unit **15** is supported rotatably around the swing axis **8** with respect to the cleaning unit **10**. The development unit **15** is urged to the cleaning unit **10** by a pressurizing spring **19***a* and a pressurizing spring **19***b*, which are elastic members, and the development roller **16** is brought into contact with the photosensitive drum **11**.

(55) Subsequently, a separation mechanism **100** which performs a contact/separation operation of the development unit **15** with respect to the cleaning unit **10** will be explained by using FIGS. **10**A and **10**B. FIGS. **10**A and **10**B are shown without the side cover **7** in order to illustrate the separation mechanism **100** provided in the printer main body P.

(56) As shown in FIG. **10**A, a protruding portion **5***b* is provided on the bearing member **5**. At a position where the protruding portion **5***b* is not in contact with the separation mechanism **100** as in FIG. **10**A, the development roller **16** is in contact with the photosensitive drum **11**. This state is an image forming position where the electrostatic latent image formed on the surface of the photosensitive drum **11** by the development roller **16** is developed.

(57) As shown in FIG. **10**B, when the separation mechanism **100** provided in the printer main body P is brought into contact with the protruding portion **5***b* and receives a force, the development unit **15** is rotated in an R**2** direction with the swing axis **8** as a rotation center, and the photosensitive drum **11** and the development roller **16** are separated. This state is a non-image forming position retreated from the image forming position.

(58) When the separation mechanism **100** returns to the original position, it is separated from the protruding portion **5***b*, and the development roller **16** and the photosensitive drum **11** are in contact again as in FIG. **10**A by the pressurizing spring **19***a* and the pressurizing spring **19***b*.

(59) As described above, the contact position (image forming position) and the separation position (non-image forming position) of the process cartridge B can be switched (movable) by the separation mechanism **100**. As a result, an attitude of the development unit **15** in the process cartridge B can be switched between the contact position and the separation position with respect to the photosensitive drum **11**. As a result, deterioration of the toner can be suppressed, and unnecessary toner consumption at non-image forming can be suppressed.

(60) As shown in FIGS. **9**A and **9**B, the cleaning unit **10** is constituted by the cleaning frame body **20**, the stay **21**, and the side cover **7**. The cleaning frame body **20** supports the cleaning blade **17**, the charging roller **12**, and the charging roller cleaner **14**. The photosensitive drum **11** is rotatably supported on one side by a drum pin **22** mounted on the cleaning frame body **20** and on the opposite side by a photosensitive-drum supporting portion **7***b* provided on the side cover **7**.

(61) Outline of Toner Cartridge C

(62) The toner cartridge C will be explained by using FIGS. **6**, **7**, **8**, **13**A, and **13**B. FIG. **6** is a front view illustrating an outline of the toner cartridge C. FIG. **7** is a sectional view (a-a section in FIG. **6**) illustrating a toner supply portion **30** of the toner cartridge C. FIG. **8** is a sectional view (b-b section in FIG. **6**) illustrating the waste-toner recovering portion **40** of the toner cartridge C. FIG. **13**A is an exploded perspective view of the toner cartridge C seen from the side (one end side) where the drive from the printer main body P is transmitted to the toner cartridge C. FIG. **13**B is an exploded perspective view of the toner cartridge C seen from the side opposite to FIG. **13**A (the other end side) with respect to the axial direction of the photosensitive drum **11**.

(63) As shown in FIGS. **6**, **7**, and **8**, the toner cartridge C has the toner supply portion **30** which supplies the toner to the process cartridge B and the waste-toner recovering portion **40** which recovers the waste toner from the process cartridge B.

(64) An outline of the toner supply portion **30** will be described below. As shown in FIGS. **7**, **8**, **13**A, and **13**B, a toner accommodating portion **30***a* of the toner supply portion **30** is formed by a supply-portion frame body **31** and a supply-portion lid **32**. The supply-portion frame body **31** has a toner discharge port **31***a* which discharges the toner from the toner accommodating portion **30***a* and

a shutter member **34** which opens/closes the toner discharge port **31***a* by rotating in an R**1** direction by interlocking with attachment of the toner cartridge C on an outside of the supply-portion frame body **31**.

(65) The toner accommodating portion **30***a* has a screw member **35** which conveys the toner toward the toner discharge port **31***a* and a stirring conveyance unit **36** which conveys the toner toward the screw member **35**.

(66) The toner having been conveyed to the toner discharge port **31***a* is discharged by volume variation of a pump **37***a*. Specifically, as shown in FIG. **13**A, it has a pump unit **37**. The pump unit **37** has the pump **37***a* which changes the volume by expansion/contraction, a cam **37***b* which expands/contracts the pump **37***a* by rotation, and a link arm **37***c*.

(67) Moreover, a drive configuration of the toner supply portion **30** will be explained by using FIG. **13**A. The toner supply portion **30** has a drive input portion **38** (first drive-input portion) which drives the conveyance unit **36** and a drive input portion **39** (second drive-input portion) which drives the pump unit **37** and the screw member **35**.

(68) Subsequently, an outline of the waste-toner recovering portion **40** will be described below. As shown in FIG. **8**, the waste-toner recovering portion **40** has a frame body **41** and an accommodating lid **42** provided on an upper part of the frame body **41** with respect to the gravity direction. In the accommodating lid **42**, a waste-toner accepting port **42***a* is provided. Moreover, the waste-toner recovering portion **40** has a shutter member **43** which opens/closes the waste-toner accepting port **42***a*. The shutter member **43** is opened/closed in an R**3** direction by interlocking with attachment/detachment of the toner cartridge C with respect to the printer main body P.

(69) As shown in FIG. **13**A, in the waste-toner recovering portion **40**, a partition member **46**, a first screw **44**, and a second screw **47** are disposed. The first screw **44** conveys the waste toner falling from the waste-toner accepting port **42***a* in the axial direction of the photosensitive drum **11**. The second screw **47** rotates by receiving a drive force from the rotation of the first screw **44** and conveys the waste toner conveyed by the first screw **44** diagonally upward relative to the gravity direction.

(70) The drive of the waste-toner recovering portion **40** is obtained as described below. As shown in FIGS. **13**A and **13**B, the drive force input by the drive input portion **38** described above is transmitted to a non-drive side of the toner supply portion **30** through the conveyance unit **36** and is transmitted to a gear **38***a*. The drive force transmitted to the gear **38***a* is transmitted to the first screw **44** through a gear train **45**.

(71) On the toner supply portion **30** side (drive side), the side cover **50** is mounted, while on the waste-toner recovering portion **40** side (non-drive side), the side cover **60** is mounted, respectively. In the configuration of this Embodiment, as also shown in FIGS. **10**A and **10**B, the drive input portion **38** and the drive input portion **39** are driven, respectively, by the drive force from a drive source provided in the printer main body P.

(72) By means of the configuration as above, even when the drive input portion **39** is not driven, the drive input portion **38** can be driven. That is, even w % ben the toner supply portion **30** is not replenishing the toner to the process cartridge B, a state where the waste toner can be recovered can be maintained by driving the first screw **44** and the second screw **47** in the waste-toner recovering portion **40**.

(73) Moreover, since the input of the drive from the printer main body P can be disposed on the one end side (drive side) of the toner cartridge C, the gear train of the printer main body P can be simplified. Furthermore, by transmitting the drive from the one end side to the other end side of the supply-portion frame body **31** by using the conveyance unit **36**, the drive can be transmitted to the waste-toner recovering portion **40** without increasing components for drive transmission. As a result, while size increase of the toner cartridge C and the printer main body P with respect to the axial direction of the photosensitive drum **11** by the drive means of the toner cartridge C is suppressed, the waste toner can be accommodated in the toner cartridge C.

(74) Attaching/Detaching Method of Process Cartridge B, Toner Cartridge C
(75) Subsequently, an attaching/detaching method of the process cartridge B, the toner cartridge C with respect to the printer main body P will be explained by using FIGS. **11**A, **11**B, and **12**A to **12**C. FIGS. **11**A and **11**B are schematic perspective views for explaining attachment of the process cartridge B, the toner cartridge C to the printer main body P. FIGS. **12**A to **12**C are schematic side views for explaining attachment of the process cartridge B, the toner cartridge C to the printer main body P.
(76) As shown in FIG. **11**A, a space inside the printer main body P is an attaching portion for the process cartridge B, the toner cartridge C. A door **107** capable of being opened/closed is provided rotatably around a rotation axis R**5** with respect to the printer main body P, and FIG. **11**A is a diagram of a state in which the door **107** is open.
(77) Moreover, the printer main body P has a guide portion **108**, a guide portion **109**. In the process cartridge B, as shown in FIGS. **9**A and **9**B, an upper boss **93**, an upper boss **94**, a lower boss **95**, and a lower boss **96** are provided on both right and left.
(78) First, the process cartridge B is attached to the printer main body P. As shown in FIGS. **11**A and **12**A, by causing the upper boss **93** (FIG. **9**B) and the lower boss **95** (FIG. **9**B) to sandwich the guide portion **108** and by causing the upper boss **94** and the lower boss **96** to sandwich the guide portion **109**, respectively, the process cartridge B inserted in an arrow direction D is attached to the printer main body P while being guided by the guide portions.
(79) The toner cartridge C has, as shown in FIGS. **13**A and **13**B, a positioning boss **50***a* and a positioning boss **60***a* on the front in an attaching direction and guided portions **50***b*, **60***b* closer to a downstream side in the attaching direction than the positioning boss **50***a*, the positioning boss **60***a*. The process cartridge B has, as shown in FIGS. **9**A and **9**B, a positioning portion **21***a* and a positioning portion **21***b* on the stay **21**.
(80) As shown in FIGS. **11**B and **12**B, the guided portion **50***b*, the guided portion **60***b* are loaded on the guide portion **108**, the guide portion **109**, respectively, and inserted in the arrow direction D.
(81) As shown in FIG. **12**C, when the toner cartridge C has been attached to an insertion completion position, the positioning boss **50***a* and the positioning boss **60***a* (FIGS. **13**A and **13**B) enter the positioning portion **21***a* and the positioning portion **21***b* (FIGS. **9**A and **9**B), respectively. At this time, distal end sides in the insertion direction of the guided portion **50***b* and the guided portion **60***b* are separated from the guide portion **108** and the guide portion **109**, respectively, and are brought into a state where rear ends are in contact with the guide portion **108** and the guide portion **109**, respectively. Moreover, the toner cartridge C is positioned to the process cartridge B by the positioning boss **50***a* and the positioning boss **60***a*, and rotating direction components around the positioning boss **50***a* and the positioning boss **60***a* of the toner cartridge C are regulated by being positioned by rear ends of the guided portion **50***b* and the guided portion **60***b*, respectively. As a result, the toner cartridge C is positioned with respect to the process cartridge B. Moreover, the rear ends of the guided portion **50***b* and the guided portion **60***b* are brought into contact with the guide portion **108** and the guide portion **109** and thus, the position of the toner cartridge B in the printer main body P is determined.
(82) When the door **107** is closed after the process cartridge B and the toner cartridge C are inserted into the printer main body P, a state where an image can be formed is brought about. An operation of removing the toner cartridge C, the process cartridge B is performed with a procedure opposite to the above.
(83) Toner Replenishing Path of Process Cartridge B
(84) Subsequently, a toner replenishing path of the process cartridge B will be explained by using FIGS. **14**, **15**A, and **15**B. FIG. **14** is a sectional perspective view illustrating a replenishing port of the process cartridge B. FIG. **15**A is a schematic sectional view for explaining a disposition relationship of the toner accepting chamber **153** and the stay **21** when the development unit **15** is positioned at the contact position with respect to the photosensitive drum **11**. Moreover, FIG. **15**B

is a schematic sectional view illustrating the disposition relationship of the toner accepting chamber **153** and the stay **21** when the development unit **15** is positioned at the separation position with respect to the photosensitive drum **11**.

(85) As described above, the process cartridge B receives toner replenishment from the toner cartridge C. Specifically, as shown in FIGS. **14**, **15**A, and **15**B, the replenishing port **21***c* for accepting the toner conveyed from the toner cartridge C is provided in the stay **21**. The toner accepted at the replenishing port **21***c* is conveyed from an accepting port **153***a* of the development unit **15** to the toner accepting chamber **153** through the delivery port **21***d*. The toner replenished to the toner accepting chamber **153** is replenished by a conveyance member **153***b* to the developer accommodating chamber **152** (FIG. **5**) from an accepting port **152***a* provided in the developer accommodating chamber **152**. A seal member **153***c* is bonded to a periphery of the accepting port **153***a*, and the seal member **153***c* seals a space between the delivery port **21***d* and the accepting port **153***a*.

(86) As shown in FIG. **15**A, shapes of the periphery of the accepting port **153***a* of the toner accepting chamber **153** and the periphery of the delivery port of the stay **21** have arc shapes R**4** around the swing axis **8** of the development unit **15** described above.

(87) As shown in FIG. **15**B, even if the development unit **15** moves to the separation position, since the replenishing port **21***c* for accepting the toner from the toner cartridge C is provided in the stay **21**, the position does not change. Moreover, a crush amount of the seal member **153***c* accompanying movement of the development unit **15** rarely changes, but stable sealing performance can be ensured regardless of the position of the development unit **15**. Furthermore, even in a state where the development unit **15** is positioned at the separation position, the accepting port **153***a* is configured to have such a size that parts of the delivery port **21***d* and the accepting port **153***a* can communicate. As a result, the toner can be accepted whichever position of the contact position (first position) and the separation position (second position) the development unit **15** is located.

(88) As shown in this Embodiment, the accepting port **153***a* may be configured to completely communicate with the delivery port **21***d* or may have a configuration of communication at least a part thereof.

(89) Configuration of Toner Accepting Chamber of Development Unit

(90) The toner accepting chamber **153** of the development unit **15** will be explained by using FIG. **16**. FIG. **16** is a schematic perspective view of the toner accepting chamber **153** of the process cartridge B.

(91) The toner cartridge C replenishes the toner to the process cartridge B by using the pump **37***a* as described above. That is, the toner replenished to the process cartridge B is replenished in a state of being mixed with air. Thus, as shown in FIG. **16**, the toner accepting chamber **153** has a downstream-side filter **153***f* and an upstream-side filter **153***g*, which are filter members for extracting air.

(92) Moreover, the conveyance member **153***b* which conveys the toner replenished into the toner accepting chamber **153** to the accepting port **152***a* is provided. The conveyance member **153***b* has a sheet member **153***d* which conveys the toner fallen from the accepting port **153***a* to the accepting port **152***a* and a spiral portion **153***e* having a shape for returning the toner from the accepting port **153***a* in a direction of the accepting port **152***a*.

(93) Regarding the filter member, the upstream-side filter **153***g* disposed on an upstream side in a conveyance direction of the spiral portion **153***e* and the downstream-side filter **153***f* disposed on an upper part of the sheet member **153***d* are disposed closer to an upper side in the gravity direction than the spiral portion **153***e*, respectively.

(94) The toner replenished from the accepting port **153***a* is conveyed by the spiral portion **153***e* and the sheet member **153***d* to the accepting port **152***a* side. That is, regarding a surface opposed to a region where the toner is conveyed by the spiral portion **153***e*, which is a lower surface of the

upstream-side filter **153***g*, since the spiral portion **153***e* is not filled with the toner due to sequential conveyance of the toner, air can be extracted stably. The downstream-side filter **153***f* is preferably disposed at least partially closer to the upper side in the gravity direction than the accepting port **153***a*. As a result, air can be extracted more efficiently when the toner is replenished to the accepting port **153***a*.
(95) Waste-Toner Conveyance Configuration Provided on Cleaning Unit **10**
(96) A waste-toner conveyance configuration provided on the cleaning unit **10** will be explained by using FIGS. **3**, **4**, **17**, **18**, and **23**. FIG. **17** is a schematic sectional view when the cleaning unit **10** is seen from an upper surface. FIG. **18** is a schematic perspective view of the process cartridge B illustrating the drive configuration of the first conveyance member **70**. Note that FIG. **18** is shown by excluding the side cover **7** and the bearing member **5** from the process cartridge B for explanation. Moreover, FIG. **23** is a schematic sectional view for explaining a positional relationship of the cleaning blade **17** and the second toner conveyance path **10***c*.
(97) As described above, the waste toner removed by the cleaning blade **17** from the photosensitive drum **11** is recovered in the waste-toner accommodating portion **10***a* as a developer recovering portion.
(98) In the waste-toner accommodating portion **10***a*, as shown in FIGS. **3** and **17**, the first conveyance member **70** is disposed with a rotation axis L**1** substantially in parallel with the rotation axis **11***b* of the photosensitive drum **11**. When the waste toner accumulates in the waste-toner accommodating portion **10***a*, the waste toner is sent out to the first toner conveyance path **10***b*. As shown in FIGS. **4** and **18**, a second conveyance member **71** is provided on the second toner conveyance path **10***c* disposed by crossing the rotation axis **11***b* of the photosensitive drum **11** (typically at a right angle). That is, the second conveyance member **71** conveys the toner in a direction different from that of the first conveyance member **70** (crossing direction). Here, the second conveyance member **71** conveys the toner upward from below an upper end of the cleaning blade with respect to the gravity direction.
(99) In this Embodiment, the second toner conveyance path **10***c* and the second conveyance member **71** are disposed on an inner side from an end part of the cleaning blade **17** in a direction (longitudinal direction) of the rotation axis **11***b* of the photosensitive drum **11** as shown in FIG. **23**. Here, the “inner side from the end part” more specifically means the inner side from the end part of the blade portion **17***a* in the cleaning blade **17**. Moreover, in the configuration of this Embodiment, the second toner conveyance path **10***c* and the second conveyance member **71** are disposed inside an end part of a development opening **151***a* in the longitudinal direction. Moreover, the second conveyance member **71** has an upstream end part and a downstream end part with respect to a conveyance direction of the developer, and the downstream end part is disposed above the upstream end part with respect to the gravity direction.
(100) The waste toner having been sent to the first toner conveyance path **10***b* is conveyed by a spiral portion **70***a* provided in the first conveyance member **70** in the direction L**1** substantially in parallel with the rotation axis direction of the photosensitive drum **11**. Then, the waste toner conveyed by the first conveyance member **70** reaches the second toner conveyance path **10***c*.
(101) Subsequently, the waste toner having reached the second toner conveyance path **10***c* is conveyed by a spiral portion **71***a* of the second conveyance member **71** as shown in FIGS. **4** and **18** in a direction L**2** above the cleaning blade **17** and orthogonal to the rotation axis **11***b* of the photosensitive drum **11**. Moreover, the waste toner having been conveyed to the end part of the second conveyance member **71** is discharged to the waste-toner accepting port **42***a* (FIGS. **13**A and **13**B) of the waste-toner recovering portion **40** from a waste-toner discharge port **72**.
(102) Here, in the image forming apparatus to which the process cartridge can be detachably attached, a size reduction of the process cartridge is in demand for the purposes of resource saving, reduction in manufacturing and conveyance costs, facilitation of handling and the like.
(103) In this Embodiment, with respect to the longitudinal direction of the process cartridge B, the

waste toner is conveyed by the configuration in which the second toner conveyance path **10***c* and the second conveyance member **71** are provided on the inner side of the end part of the cleaning blade **17**. As a result, a length in the longitudinal direction of the process cartridge B can be made shorter. That is, the size of the process cartridge B in a direction of the rotation axis **11***b* of the photosensitive drum **11** can be reduced. Moreover, in order to make conveyance performance of the waste toner more stable, a pitch of the spiral portion **71***a* of the second conveyance member **71** may be made longer than a pitch (distance between adjacent spiral portions) of the spiral portion **70***a* of the first conveyance member **70**. Similarly, in order to make the conveyance performance of the waste toner more stable, a diameter of the pitch of the spiral portion **71***a* may be made larger than a diameter of the pitch of the spiral portion **70***a*.

(104) Drive Configuration of Waste-Toner Conveyance Member

(105) The drive configurations of the first conveyance member **70**, the second conveyance member **71** will be explained by using FIG. **18**. FIG. **18** is a schematic perspective view of the process cartridge B illustrating the drive configuration of the waste-toner conveyance member. Note that it is an explanatory view excluding the side cover **7** and the bearing member **5** from the process cartridge B for explanation.

(106) As shown in FIG. **18**, in the cleaning unit **10**, a first gear **70***b*, which is a gear for transmitting drive to the first conveyance member **70**, and a second gear **71***b*, which is a bevel gear for transmitting the drive to the second conveyance member **71** are disposed. To the first gear **70***b*, the drive is transmitted from a development coupling **155** provided coaxially with the swing axis **8** through a gear train **601**.

(107) Moreover, the drive is transmitted to the second gear **71***b* from the first gear **70***b* through an idler gear **602**, a drive gear **75***b* provided on a through shaft disposed on a drive side, a through shaft **75** disposed in parallel with the first conveyance member **70**, and a bevel gear **75***a* provided on an end part on a non-drive side of the through shaft **75**. As a result, the drive is transmitted to the first conveyance member **70**, the second conveyance member **71**, and the waste toner can be conveyed. Moreover, though the bevel gear is used in this Embodiment, the second gear **71***b* and the bevel gear **75***a* may be screw gears. Furthermore, in this Embodiment, rotation numbers of the first conveyance member **70** and the second conveyance member **71** are the same, but in order to improve conveyance efficiency, the rotation number of the second conveyance member **71** may be higher than that of the first conveyance member **70**.

Embodiment 2

(108) Subsequently. Embodiment 2 will be explained. For the same configuration as that in Embodiment 1, explanation will be simplified.

(109) Waste-Toner Conveyance Configuration of Waste-Toner Accommodating Portion **10***a*

(110) A configuration which makes the waste-toner conveyance of the waste-toner accommodating portion **10***a* in Embodiment 1 more stable will be explained by using FIGS. **19**, **20**, and **21**. FIG. **19** is a sectional view illustrating an outline of the process cartridge B in Embodiment 2, FIG. **20** is a sectional view illustrating the second toner conveyance path **10***c* of the process cartridge B in Embodiment 2, and FIG. **21** is a schematic sectional view when the cleaning unit **10** is seen from an upper surface.

(111) As shown in FIGS. **19**, **20**, and **21**, the waste-toner accommodating portion **10***a* has a third conveyance member **73** and a first sheet portion **74**, and the third conveyance member **73** has a spiral portion **73***a* and a fixing portion **73***b* for fixing the first sheet portion **74**.

(112) As described above, the waste toner on the photosensitive drum **11** is recovered by the cleaning blade **17** in the waste-toner accommodating portion **10***a*. The recovered waste toner is conveyed by the spiral portion **73***a* of the third conveyance member **73** toward a center in the rotation axis **11***b* direction of the photosensitive drum **11**. The waste toner having been conveyed to the center in the rotation axis **11***b* direction of the photosensitive drum **11** is conveyed by the first sheet portion **74** to the first toner conveyance path **10***b* provided in a direction orthogonal to the

rotation axis **11***b* direction of the photosensitive drum **11**.
(113) The waste toner having reached the first toner conveyance path **10***b* is conveyed by the spiral portion **70***a* provided in the first conveyance member **70** in the rotation axis **11***b* direction of the photosensitive drum **11**. Subsequently, the waste toner having been conveyed by the first conveyance member **70** reaches the second toner conveyance path **10***c*. The waste toner having reached the second toner conveyance path **10***c* is conveyed by the spiral portion **71***a* of the second conveyance member **71** in a direction above the cleaning blade **17** and orthogonal to the rotation axis **11***b* of the photosensitive drum **11**.
(114) Moreover, the waste toner conveyed to the end part of the second conveyance member **71** is discharged to the waste-toner accepting port **42***a* (FIG. **13**A) of the waste-toner recovering portion **40** from the waste-toner discharge port **72**. By conveying the waste toner in the configuration as above, collection of the waste toner in the waste-toner accommodating portion **10***a* (particularly the second toner conveyance path **10***c* in the rotation axis **11***b* direction of the photosensitive drum **11**) can be prevented, and stable conveyance of the waste toner is made possible. Moreover, a drive load can be also reduced.
(115) Subsequently, another configuration of the configuration of the conveyance by the spiral portion **73***a* of the third conveyance member **73** toward the center in the rotation axis **11***b* direction of the photosensitive drum **11** described above will be explained by FIGS. **28**, **29**, **30**A, and **30**B.
(116) FIG. **28** is a schematic sectional view of the cleaning unit **10** when seen from the upper surface, FIG. **29** is a front view of the third conveyance member **73** unit, and FIGS. **30**A and **30**B are sectional views of the third conveyance member **73** unit. FIG. **30**A shows an a-a section of FIG. **29**, and FIG. **30**B for a b-b section.
(117) As shown in FIGS. **28** and **29**, a fixing portion **73***d*, a fixing portion **73***e* are provided for fixing the second sheet portion **701** and the third sheet portion **702** on both end parts of the third conveyance member **73**. The second sheet portion **701** and the third sheet portion **702** have slits **701***c*, **702***c* from free ends **701***a*, **702***a* toward fixed ends **701***b*, **702***b*. Explanation on the slits **701***c*, **702***c*, which will be described later, will be made for the second sheet portion **701**, since they have symmetrical shapes. The slit **701***c* from the fixed end **701***b* toward the free end **701***a* of the second sheet portion **701** is inclined from the end part in the rotation axis **11***b* direction of the photosensitive drum **11** toward the center. By means of the drive configuration described later, when the third conveyance member **73** is rotated, the waste toner recovered in the waste-toner accommodating portion **10***a* can be conveyed by the second sheet portion **701** and the third sheet portion **702** toward the center in the rotation axis **11***b* direction of the photosensitive drum **11**. The toner can be conveyed in a state where density of the waste toner of the waste-toner accommodating portion **10***a* is lower than that in the configuration of the spiral portion **73***a*.
(118) Moreover, as shown in FIGS. **28**, **29**, **30**A, and **30**B, a relationship between a sectional area S**1** of the fixing portion **73***d* of the third conveyance member **73** and a sectional area S**2** of the fixing portion **73***e* is set to S**1**>S**2**. Since S**1** is on a drive input side, distortion of the third conveyance member **73** can be prevented, and the waste toner can be conveyed stably. Moreover, since the sectional areas are in the relationship of S**1**>S**2**, free lengths of the second sheet portion **701** and the third sheet portion **702** are different. By changing angles of the slits **701***c*, **702***c*, rigidity of the second sheet portion **701** and the third sheet portion **702** are made equal. As a result, the conveyance force of the waste toner in the rotation axis **11***b* direction of the photosensitive drum **11** can be made equal, and the waste toner can be conveyed stably.
(119) Moreover, as shown in FIGS. **28** and **29**, a fixed phase of the first waste-toner conveyance sheet **74** with respect to the third conveyance member **73** is shifted from the fixed phases of the second sheet portion **701**, the third sheet portion **702**. As a result, the waste toner conveyed by the second sheet portion **701** and the third sheet portion **702** toward the center in the rotation axis **11***b* direction of the photosensitive drum **11** can be efficiently conveyed by the first waste-toner conveyance sheet **74** to the first toner conveyance path **10***b*.

(120) Drive Configuration of Waste-Toner Conveyance Member
(121) The drive configuration of the third conveyance member **73** will be explained by using FIG. **22**. FIG. **22** is a schematic perspective view of the process cartridge B illustrating the drive configuration of the waste-toner conveyance member.
(122) As shown in FIG. **22**, in the cleaning unit **10**, a third gear **73***c*, which is a gear which transmits drive to the third conveyance member **73** is disposed. To the third gear **73***c*, the drive is transmitted from the development coupling **155** provided coaxially with the swing axis **8** through the gear train **601**. Similarly, to the first gear **70***b*, the drive is transmitted from the third gear **73***c* through an intermediate gear **603**. Moreover, to the second gear **71***b*, the drive is transmitted from the first gear **70***b* through the idler gear **602**, the drive gear **75***b*, the through shaft **75** disposed in parallel with the first conveyance member **70**, and the bevel gear **75***a* provided at the end part of the through shaft **75**. As a result, the drive is transmitted to the first conveyance member **70**, the second conveyance member **71**, the third conveyance member **73**, and the waste toner can be conveyed.
Embodiment 3
(123) Subsequently, Embodiment 3 will be explained. For the same configuration as those in Embodiments above, explanation will be simplified.
(124) FIG. **24** is a schematic sectional view illustrating a positional relationship of the first conveyance member **70** and the second conveyance member **71**. FIG. **25** is a schematic sectional view passing through axes of the first conveyance member **70** and the second conveyance member **71**. FIG. **25** illustrates a state of the first conveyance member **70** when seen from one side in the longitudinal direction.
(125) Here, as described above, the second conveyance member **71** needs to convey the waste toner against the gravity direction. However, the waste toner having dripped down from a gap between the second toner conveyance path **10***c* and the spiral portion **71***a* of the second conveyance member **71** backflows through the second toner conveyance path **10***c* to the upstream side in the waste-toner conveyance direction of the second conveyance member **71**, that is, toward the lower side in the gravity direction and thus, a flow in a direction opposite to the L**2** direction (FIG. **22**) is generated. The backflowing waste toner reaches the first toner conveyance path **10***b* via the second toner conveyance path **10***c* and is pushed back again by the first conveyance member **70** to the side of the second conveyance member **71**. Thus, the first conveyance member **70** preferably has such configuration of continuously pushing the waste toner in the first toner conveyance path **10***b* into the second toner conveyance path **10***c*. That is, when the waste toner conveyed by the spiral portion **70***a* of the first conveyance member **70** is to be sent out toward the spiral portion **71***a* of the second conveyance member **71**, it is preferable that a stable powder pressure is continuously applied.
(126) In this Embodiment, as shown in FIGS. **24** and **25**, an axial-direction projected surface **76** of the spiral portion **70***a* in the first conveyance member **70** is disposed so as to overlap the spiral portion **71***a* of the second conveyance member **71**. Moreover, in order to prevent contact between the first conveyance member **70** and the second conveyance member **71**, the first conveyance member **70** is disposed with a gap from an outer surface of the spiral portion **71***a* in the second conveyance member **71**. Furthermore, a crossing portion **10***bi* with the second toner conveyance path **10***c* has a tunnel shape surrounding the whole outer peripheral surface of the spiral portion **70***a* of the first conveyance member **70**.
(127) By having the configuration as above, the powder pressure of the waste toner conveyed by the spiral portion **70***a* can be continuously applied to the spiral portion **71***a* stably. Moreover, even if the insides of the first toner conveyance path **10***b* and the second toner conveyance path **10***c* are filled with the waste toner, the spiral portion **70***a* and the spiral portion **71***a* are configured such that a conveyance amount of the second conveyance member **71** is larger than that of the first conveyance member **70** so that the waste toner does not stagnate. As described above, it is configured such that a load rise for the first conveyance member **70** to push back the backflow

waste toner is suppressed even when the waste toner backflows, and a drive-force rise of the first conveyance member **70**, which is a conveyance force of the waste toner, can be suppressed.

(128) In this Embodiment, as shown in FIGS. **24** and **25**, it is configured such that the axes of the first conveyance member **70** and the second conveyance member **71** are on the same plane, but the axes do not have to be on the same plane. In this case, by configuring such that the axial-direction projected surface of the spiral portion **70***a* of the first conveyance member **70** overlaps a part of the second conveyance member **71** and by disposing the first conveyance member **70** closer to the upper side in the gravity direction of the second conveyance member **71**, the backflow toner intruding into the first toner conveyance path **10***b* can be suppressed.

(129) Regarding the drive configuration of the waste-toner conveyance member explained by referring to FIG. **18**, in this Embodiment, the through shaft **75** is exposed to an outside of the cleaning unit **10** and is provided on an opposite side from the photosensitive drum **11** through the waste-toner accommodating portion **10***a*. It is configured such that the drive transmitting members of the first conveyance member **70** and the second conveyance member **71** are disposed outside the waste-toner accommodating portion **10***a*, the first toner conveyance path **10***b*, and the second toner conveyance path **10***c* so that the waste toner does not adhere to the drive transmitting member. By having the configuration as above, reduction of drive transmission efficiency is suppressed, the drive-force rise of the waste-toner conveyance is suppressed, and a stable waste-toner conveyance configuration is realized. In this Embodiment, the drive transmitting member is provided by being exposed to the outside of the process cartridge B.

Embodiment 4

(130) Subsequently, Embodiment 4 will be explained. For the same configuration as those in the Embodiments above, explanation will be simplified. Particularly in this Embodiment, the drive configuration of the waste-toner conveyance member will be explained by using FIGS. **9**, **26**, and **27**. Here, in the present invention, the cleaning unit **10** is supported capable of swing with respect to the development unit **15**, and in the configuration of contact and separation, the rotatable waste-toner conveyance member is disposed in the cleaning unit **10**. In this configuration, as described above, such a configuration is in demand that a rotary force is stably transmitted by the waste-toner conveyance member, even if the cleaning unit **10** is rotated with respect to the development unit **15** and goes back and forth between the contact state and the separation state.

(131) As shown in FIG. **27**, in the development unit **15**, the development coupling **155** as a drive input member for receiving the drive from the printer main body P is provided. The development coupling **155** is disposed substantially coaxially with the gear member **5***a* provided on the bearing member **5** (FIGS. **10**A and **10**B) and is disposed substantially coaxially with the swing axis **8**. The gear member **5***a* is rotated with the rotation of the development coupling **155**. That is, when seen from a direction of the swing axis **8**, each member is disposed so that the rotation axis of the development coupling **155**, a rotation center of the gear member **5***a*, and the swing axis **8** overlap. The development coupling **155** transmits a rotary force to a development roller gear **16***a* for transmitting the rotary force to the development roller **16**, a stirring gear **154***a* for transmitting the rotary force to the stirring member **154** and the like.

(132) Moreover, the development coupling **155** is connected to a first idler gear **601***a* disposed on an upstream side in a rotating direction of the gear train **601** as a rotary-force transmitting member provided in the cleaning unit **10** by the gear member **5***a* Furthermore, the first gear **70***b* for transmitting the rotary force to the first conveyance member **70** is disposed coaxially with the first conveyance member **70**, and the first conveyance member **70** and the first gear **70***b* are integrally and rotatably connected (FIG. **26**). Then, the gear train **601** is connected to the first gear **70***b*, and the rotary force is transmitted from the gear train **601** to the first gear **70***b*.

(133) As described above, the development unit **15** is configured to move between the contact position and the separation position by rotating around the swing axis **8**, which is substantially coaxial with the rotation axis of the development coupling **155**, with respect to the cleaning unit **10**.

Thus, when the development unit **15** moves between the contact position and the separation position with respect to the photosensitive drum **11**, a position change of the development coupling **155** can be kept very small of approximately fitting backlash.
(134) Specifically, the gear member **5***a* is engaged with the hole shape **7***a* of the side cover **7** (FIG. **19**A), while the gear member **5***a* of the bearing member **5** and the hole shape **7***a* of the side cover **7** are engaged with each other within the fitting backlash of 0.13 mm. Thus, the position of the development coupling **155** when the development unit **15** moves between the contact position and the separation position does not change more than a portion of the fitting backlash, and the position of the development coupling **155** is rarely changed as show % n in FIGS. **10**A and **10**B. Therefore, by having the configuration in which the drive force is transmitted from the development coupling **155** to the cleaning unit **10** through the gear member **5***a* and the gear train **601**, the following effects can be obtained. That is, meshing of gear pairs across units from the development unit **15** to the cleaning unit **10** can be made stable. In other words, the meshed state of the development coupling **155** provided in the development unit **15** and the first idler gear **601***a* provided in the cleaning unit **10** can be maintained. As a result, during movement between the contact position and the separation position, occurrence of damage on the development coupling **155** and the first idler gear **601***a* connected to the development coupling **155** can be reduced.
(135) As described above, in the configuration of this Embodiment, the drive force is transmitted from the development coupling **155** as the drive input member to the member provided in the cleaning unit **10** through the gear member **5***a* and the gear train **601**. The gear member **5***a* is provided in the development unit **15**, which is the second unit, and is rotated coaxially with the rotation axis of the development coupling **155** and coaxially also with the swing axis **8**. And the first gear **70***b* for rotating the first conveyance member **70** and the gear train **601** for transmitting the driver force to the first gear **70***b* are provided in the cleaning unit **10** as the second unit. By means of this configuration, even if the development unit **15** moves between the contact position and the separation position, the meshing between the gears can be maintained without disconnection of the drive transmission path to the first gear **70***b*. As in the configuration of this Embodiment, in order to maintain the meshing between the gears regardless of the swing operation of the development unit **15**, it is preferable that gears other than the inner gear member **5***a*, which is a gear for transmitting the drive to the first gear **70***b*, are all provided on the cleaning unit **10** side.
(136) In addition, this configuration is not a configuration of transmitting the rotary force from a drum gear **11***a***1** provided in the photosensitive drum **11** to the first gear **70***b* as the cleaning unit **10**. Thus, since it is configured such that a load variation or swing during rotation of the first conveyance member **70** is hardly transmitted to the photosensitive drum **11**, the photosensitive drum **11** can be rotated stably.
(137) Note that, in this Embodiment, the configuration in which the rotary force is transmitted from the development coupling **155** to the first gear **70***b* through the gear train **601** was explained, but this is not limiting. For example, such a configuration that the first gear **70***b* is connected to the development coupling **155** not through the gear train **601** can be suitably applied to the present invention.
(138) Moreover, in this Embodiment, the configuration of transmitting the rotary force from the development coupling **155** to the first conveyance member **70** as the rotating member provided in the cleaning unit **10** was explained, but this is not limiting. For example, such a configuration that a rotating member other than the photosensitive drum **11** provided in the cleaning unit **10** such as the charging roller **12** is rotated can be suitably applied to the present invention.
(139) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(140) This application claims the benefit of Japanese Patent Application No. 2022-102192, filed on

Jun. 24, 2022, and, Japanese Patent Application No. 2023-054270, filed on Mar. 29, 2023, which are hereby incorporated by reference wherein in their entirety.

Claims

1. A process unit comprising: a rotatable photosensitive drum configured to bear a developer; a development unit including a developer accommodating chamber configured to accommodate the developer supplied from a developer supply portion provided in a developer cartridge and a development roller configured to develop an electrostatic image formed on the photosensitive drum with the developer supplied from the developer accommodating chamber; a transfer roller configured to transfer a toner image developed on the photosensitive drum to a sheet; a cleaning member configured to remove the developer remaining on a surface of the photosensitive drum, the cleaning member including a contact portion in contact with a surface of the photosensitive drum and a support portion configured to support the contact portion; a developer recovering portion configured to recover the developer removed by the cleaning member; a first developer conveyance path; a second developer conveyance path; a first conveyance member disposed in the first developer conveyance path and configured to convey the recovered developer in a rotation axis direction of the photosensitive drum from the developer recovering portion to the second developer conveyance path; a second conveyance member disposed in the second developer conveyance path and configured to convey the recovered developer in a direction crossing the rotation axis direction to a discharge port in the second developer conveyance path through which the developer is conveyed to a waste-developer recovering portion provided in the developer cartridge; and a third conveyance member configured to convey the recovered developer toward a center of the photosensitive drum in the rotation axis direction, wherein, in a state that the process unit is attached to an apparatus main body of an image forming apparatus, in a section perpendicular to the axis of rotation of the photosensitive drum, (i) the transfer roller is located below the photosensitive drum, (ii) the second conveyance member is above the photosensitive drum, (iii) the second conveyance member conveys the recovered developer upward relative to a gravity direction, and (iv) the recovered developer is discharged from the discharge port to the waste-developer recovering portion, wherein in the rotation axis direction, the second conveyance member is disposed in an inner side of the contact portion of the cleaning member with respect to an end part of the contact portion of the cleaning member, and wherein the developer recovered by the cleaning member is conveyed by the third conveyance member and is then conveyed by the first conveyance member.
2. The process unit according to claim 1, wherein each of the first conveyance member and the second conveyance member has a spiral portion configured to convey the developer.
3. The process unit according to claim 2, wherein a pitch of the spiral portion of the second conveyance member is longer than a pitch of the spiral portion of the first conveyance member.
4. The process unit according to claim 2, wherein a diameter of the spiral portion of the second conveyance member is larger than a diameter of the spiral portion of the first conveyance member.
5. The process unit according to claim 1, wherein the third conveyance member has a sheet member configured to convey the developer to the first conveyance member.